

Collective Cognitive Security by Design: What Makes Human Groups Cognitively Secure and Agentic in AI-Rich Environments?

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Generative and agentic AI are becoming embedded in the cognitive infrastructure of work, education, and decision-making. As AI increasingly participates in framing problems, generating alternatives, evaluating options, and shaping decisions, the cognitive-security challenge extends beyond inaccurate or manipulative information. Even when AI performs well, substitution of human cognitive work may reduce opportunities for people to retrieve knowledge, generate alternatives, question assumptions, exercise judgment, and retain ownership of decisions. It may also reduce opportunities for idiosyncratic human knowledge, including experience-based, contextual, minority, and locally held perspectives, to enter collective reasoning before a shared AI-generated frame organizes the problem.

We use Collective Cognitive Security (CCS) to denote the capacity of a group to preserve the conditions under which its members can independently generate, challenge, integrate, own, and act on judgments while interacting with AI. We distinguish CCS from collective intelligence: a group may produce a correct or high-quality outcome while remaining cognitively insecure if its members no longer meaningfully generate, evaluate, challenge, or own the reasoning that produced it.

We further propose the Epistemic Agency Practice Loop (EAPL) as a candidate mechanism through which epistemic agency may be repeatedly exercised, and Designed Collective Thinking as a candidate social architecture for sustaining participation. We outline a research agenda examining how deliberation structure and AI-entry timing affect enacted agency, cognitive diversity, idiosyncratic knowledge expression, decision ownership, responsibility, idea provenance, AI reliance, and cognitive-capability retention.

The broader question is how human-AI systems can be designed so that augmentation does not erode the human capacities, expertise, and collective knowledge on which resilient organizations and societies depend.

Additional Key Words and Phrases: Cognitive Security, Collective Cognitive Security, Epistemic Agency, Human-AI Interaction, Collective Intelligence, Cognitive Diversity, Idiosyncratic Knowledge

1 Introduction: The New AI-Rich Cognitive Playing Field

Generative and increasingly agentic AI systems are becoming part of the cognitive infrastructure of professional and educational life. They do not merely retrieve information. They help frame problems, generate alternatives, summarize evidence, evaluate options, draft arguments, and recommend actions. This creates substantial opportunities for augmentation, but it also changes the conditions under which human judgment is formed.

Much AI-risk research appropriately addresses inaccurate outputs, misinformation, bias, manipulation, and inappropriate reliance. Yet a complementary cognitive-security problem may arise even when AI performs well. If a fluent AI-generated response enters reasoning early, people may have fewer opportunities to retrieve what they know, formulate independent interpretations, surface uncertainty, generate alternatives, or articulate experience-based knowledge before a shared frame is introduced. Interaction design already matters for reliance and cognition: cognitive-forcing interventions can reduce overreliance on AI [5], while human-led and model-led architectures can differentially affect critical-thinking effort, responsibility, diversity, and ownership [13, 14, 16, 19]. A large meta-analysis further shows that human-AI synergy is heterogeneous rather than automatic [22].

The deeper concern is therefore not only whether AI may provide a wrong answer, but whether repeated cognitive substitution reduces the exercise of the practices through which humans remain capable epistemic agents: retrieving, questioning, evaluating, comparing, judging, choosing, and taking responsibility. Emerging work on human cognitive

security and epistemic agency provides a foundation for this question [1, 7, 24]. The issue becomes especially consequential in groups, where collective reasoning depends on whether distributed perspectives and uniquely held information are surfaced and integrated. Groups systematically overweight shared information relative to unique information [15], while collective performance depends on interaction patterns rather than merely individual ability [20, 23].

A longer-term implication concerns expertise. Automation can reduce opportunities for deliberate practice and progressively complex work [3]; case evidence shows how reliance on cognitive automation can generate self-reinforcing skill erosion [21]; and a recent review identifies AI-induced deskilling and inhibited skill acquisition as emerging concerns in medicine [17]. These studies do not establish inevitable decline, but they motivate a developmental question: if humans increasingly delegate problem framing, reasoning, and evaluation, how will future human expertise continue to develop?

We therefore ask: **What makes human groups cognitively secure and agentic in AI-rich environments?** We use **Collective Cognitive Security (CCS)** to denote the capacity of a group to preserve the human cognitive and social conditions required for its members to generate, challenge, integrate, own, and act on judgments while interacting with AI. Our central proposition is simple: **a group can be correct without being cognitively secure.**

2 From Cognitive Security to Collective Cognitive Security

Cognitive security remains a heterogeneous field. Early work framed cognition security around the effects of fake news on perception, knowledge, attitudes, and decisions [11]. More recent accounts broaden the construct toward resilience against malign influence and toward the protection of cognitive assets, including human cognition and organizational decision structures [6, 10]. Allred et al. take an important operational step by defining human cognitive security through observable outcomes such as veracity discernment, task-oriented action, and information sharing [1].

We extend this emerging agenda to a different failure mode. Cognitive vulnerability need not originate only from malicious or inaccurate information. It may also arise from *successful cognitive substitution*. An AI system can provide useful and accurate input while reducing the extent to which humans independently generate alternatives, retrieve their own knowledge, evaluate competing claims, or experience ownership of the resulting judgment. Outcome quality can therefore conceal reduced human epistemic participation.

At the group level, this failure mode is not reducible to individual cognition. Collective reasoning depends on whether different members contribute nonredundant knowledge, whether disagreement is possible, and whether people remain participants in producing and evaluating the collective judgment. Shared epistemic agency already demonstrates that agency can be studied as an emergent group process [8], while psychological safety concerns the interpersonal conditions for speaking and learning [9]. CCS is distinct from both. In simplified terms, psychological safety asks, *Can I speak?*; epistemic agency asks, *Can I generate and own judgment?*; collective intelligence asks, *How well can the group perform?*; CCS asks, *Can the group preserve the conditions that allow its human members to remain epistemic actors while AI participates in cognition?*

This distinction matters because high task performance does not guarantee human cognitive security. A team can immediately adopt a correct AI recommendation yet contribute little independent knowledge, fail to examine alternatives, and retain little ownership or responsibility. CCS therefore adds a protective and developmental layer to collective performance rather than replacing collective intelligence.

3 Epistemic Agency, Idiosyncratic Knowledge, and the EAPL

Epistemic agency concerns an active role in producing, evaluating, revising, and taking responsibility for knowledge and judgment. AI can influence belief formation and revision in ways that may diminish such agency [7], while recent work emphasizes strengthening active human epistemic agency in generative-AI partnerships [24]. Organizational research further suggests that agency can circulate between humans and AI through delegation, attribution, displacement, and reassertion [2, 13].

We distinguish **perceived agency** from **enacted epistemic agency**. The former concerns whether people feel autonomous or responsible. The latter concerns what they actually do: retrieve prior knowledge, formulate an independent interpretation, generate alternatives, ask questions, challenge claims, evaluate evidence, revise judgments, choose, and own a decision. This makes agency observable in interaction rather than only reportable after it.

For CCS, a particularly important form of enacted agency is the expression of **idiosyncratic human knowledge**. Group members enter reasoning with different professional experiences, local information, contextual cues, memories of exceptional cases, minority interpretations, uncertainties, and intuitions. Hidden-profile research shows that uniquely held information is systematically less likely to enter group discussion, even when it is important for decision quality [15]. In AI-rich settings, a shared AI analysis may intensify this problem by providing an immediately available common frame before nonredundant human contributions have surfaced.

We therefore propose a central design objective of CCS: **protect the opportunity for idiosyncratic human knowledge to enter collective reasoning before convergence**. The aim is not to privilege human information simply because it is human. Human reasoning is also vulnerable to bias, conformity, and error. The aim is to prevent potentially valuable distributed knowledge from disappearing before it can be expressed and evaluated.

We further introduce the **Epistemic Agency Practice Loop (EAPL)** as an integrative candidate process model:

Articulate → Retrieve → Generate → Listen → Diverge → Question → Evaluate → Integrate → Choose → Own → Commit → Act → Reflect

The EAPL is not proposed as a newly discovered psychological law. It integrates established traditions in shared epistemic agency, action cycles, professional agency, and reflective learning [8, 12]. Its proposed novelty is functional: it makes repeated enactment of epistemic agency observable and testable in AI-rich group settings. Articulation and retrieval activate human knowledge; divergence allows nonredundant contributions to surface; questioning and evaluation preserve active comparison; choice and ownership preserve responsibility; action and reflection reconnect cognition with consequences in the world. Repeated participation may therefore serve as a candidate mechanism for maintaining agency, a hypothesis to be tested rather than assumed.

4 Collective Cognitive Security by Design

If CCS depends on preserving active epistemic participation, cognitive security becomes partly a design problem. We use **Designed Collective Thinking (DCT)** to describe structured human deliberation intentionally configured to elicit independent retrieval, nonredundant knowledge, questioning, perspective-taking, dissent, evaluation, ownership, and action. DCT is not a single branded method; it is a family of deliberative architectures.

A core design principle is **Protected Human Divergence**: a bounded phase in which participants formulate and express their own interpretations before exposure to a common AI-generated analysis. This principle is consistent with evidence that the architecture of social learning changes exploration, convergence, and group performance [4]. We propose the following candidate sequence:

Proposed Research Architecture for Collective Cognitive Security

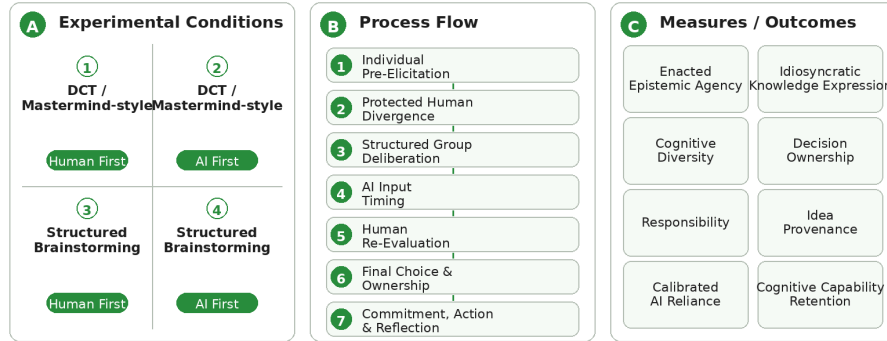


Figure 1. Proposed research architecture for testing Collective Cognitive Security (CCS). The framework links experimental conditions (deliberation architecture and AI-entry timing) to a structured human?AI process and to outcomes indexing epistemic agency, idiosyncratic knowledge expression, cognitive diversity, ownership, responsibility, idea provenance, calibrated AI reliance, and cognitive capability retention.

The purpose is not to delay AI arbitrarily or to assume that humans are intrinsically superior. Rather, Human First protects an epistemic opportunity. AI can subsequently challenge assumptions, identify missing evidence, generate alternatives, and synthesize competing positions. **Human First does not mean AI Last. It means preserving an opportunity for human cognitive diversity to become visible before a shared machine-generated frame begins to organize collective thought.**

Protected divergence is insufficient if people do not feel able to contribute. Psychological safety and high-quality listening may therefore function as enabling conditions for epistemic participation. We hypothesize that listening can matter cognitively, not only relationally, by increasing the likelihood that uncertainty, dissent, unfinished reasoning, and idiosyncratic knowledge are surfaced. This relationship remains empirical rather than assumed.

The research agenda is also **practice-informed**. It emerged partly from the author's long-term facilitation of structured peer-deliberation processes in organizational and academic settings. A recurring pattern was that participants articulated personally meaningful problems, elicited experience-based perspectives, evaluated competing contributions, retained authority over the final choice, committed to action, and returned with experience generated through implementation. These observations are not evidence of effectiveness. They generated the hypothesis that structured peer deliberation can be decomposed into testable cognitive and agentic mechanisms. One candidate implementation is a Mastermind-style protocol, treated here as a practice-derived social technology for empirical comparison rather than an assumed solution. In short, **practice generated the hypothesis; research must test the mechanism.**

5 Research Agenda

The framework leads to four linked empirical programs.

Study 1: Cognitive Security Discovery Study. A multi-team observational study would examine AI-rich teams across organizational and educational settings. Recorded meetings and surveys would capture AI-entry timing, psychological safety, listening, participation equality, cognitive diversity, ownership, responsibility, reliance, and perceived human contribution. Before collective deliberation, individual elicitation would capture each participant's initial knowledge, interpretations, and judgments, providing a baseline against which transcript coding could estimate which uniquely held contributions entered the collective reasoning process. Transcript coding would also measure enacted agency. The objective is exploratory: *what predicts Collective Cognitive Security?*

Study 2: Causal Architecture Experiment. A repeated 2×2 design would compare deliberation architecture and AI timing: (a) a DCT/Mastermind-style protocol versus structured brainstorming, crossed with (b) AI First versus Human First. Human First would require independent retrieval and human divergence before AI input; AI First would expose participants to the same AI analysis before human deliberation. Some tasks could use hidden-profile designs in which nonredundant information is experimentally distributed across group members, enabling direct measurement of whether uniquely held knowledge is surfaced and integrated. The design would test whether different architectures optimize different outcomes rather than presuming one universal winner.

Study 3: Longitudinal EAPL Study. Participants would engage in repeated structured deliberation over 8–12 weeks. The central question is whether repeated enactment of retrieval, divergence, evaluation, ownership, action, and reflection predicts changes in perceived and enacted agency, independent judgment, responsibility, and willingness to challenge AI.

Study 4: Capability-Retention and Transfer Test. After repeated AI-rich work, participants would complete a novel task with AI removed or experimentally limited. Outcomes would include alternative generation, critical evaluation, error detection, judgment quality, confidence calibration, and ability to explain reasoning. This study would directly test the strongest developmental claim: whether interaction architecture is associated with retention or transfer of human cognitive capability.

Across studies, measurement should go beyond self-report. Candidate behavioral indicators include independent ideas generated before AI exposure, unique information surfaced, dissent and challenge acts, revisions to AI suggestions, provenance of adopted ideas (human-self, human-other, group, AI, hybrid), decision explanation, ownership, commitment, and no-AI transfer performance. The framework also requires discriminant testing: CCS should explain meaningful variance beyond psychological safety, collective intelligence, and perceived agency.

The agenda maps naturally onto a cognitive-vulnerability and mitigation logic. The AI affordance is immediate shared analysis; the vulnerability is reduced independent retrieval and idiosyncratic knowledge expression; the failure pattern is shared AI anchoring and premature convergence; the candidate mitigation is protected human divergence plus structured deliberation; the metrics are agency, provenance, unique-information expression, ownership, reliance, and transfer. Burdens such as time, facilitation, hierarchy, verbal dominance, and cognitive effort should be measured rather than ignored.

6 Broader Implications and Boundary Conditions

The framework has implications beyond immediate decision quality. Human epistemic agency is part of the infrastructure through which future expertise is generated. Professional knowledge includes codified information, but also experiential, procedural, contextual, and exception-based know-how developed through action, feedback, and practice. Organizational knowledge creation likewise depends on processes through which individual knowledge is articulated and amplified [18]. If human engagement in difficult reasoning is progressively displaced rather than augmented, AI-rich organizations may become information-rich while weakening internally generated human expertise.

These implications are hypotheses, not inevitabilities. Human First may not outperform AI First on every task. Early AI input may improve speed, accuracy, or access to evidence. Human groups can also amplify bias, conformity, misinformation, and status effects. DCT may impose meaningful time and cognitive burdens. The framework is therefore falsifiable: if Human-First DCT does not increase enacted agency, idiosyncratic knowledge expression, critical evaluation, ownership, or later independent capability relative to alternatives, its proposed cognitive-security value is weakened.

The question is not whether humans should think without AI, but **which human-AI architectures preserve which human cognitive capacities, for which tasks, under which conditions, and at what cost?**

7 Conclusion

Cognitive security in AI-rich environments should include more than protection from false or manipulative content. It should also ask whether human groups preserve the conditions required to reason for themselves. Collective Cognitive Security offers a group-level lens on this problem, while EAPL and Designed Collective Thinking provide candidate mechanisms and architectures for empirical testing. The goal is not to preserve humans by excluding AI, but to design human-AI systems in which augmentation does not require the surrender of human epistemic agency. The challenge is not only to keep humans in the loop. It is to keep humans developing in the loop.

Generative AI Use Disclosure

ChatGPT (OpenAI) was used as a writing and research-assistance tool to help organize author-developed material, compare alternative formulations, support literature discovery, and improve English-language clarity. The research problem and practice-informed hypothesis originated in the author's prior professional and academic work. The author made all final conceptual and interpretive decisions, critically revised all AI-assisted text, and verified cited scholarly sources against publisher or repository records. No AI system is listed as an author, and the author assumes full responsibility for the accuracy, originality, interpretation, and integrity of the submitted work.

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